

Python Libraries

<https://docs.python.org/3/library/index.html>

EXPLORER

USING LIBRARIES

dates_demo.py

math_demo.py

matplotlib_demo.py

random_demo.py

math_demo.py

do_some_math

```
1 import statistics
2 import math
3
4 def do_some_math():
5     numbers = [1,2,3,4,5,6,7,8,9]
6
7     print(statistics.mean(numbers))
8
9     for i in numbers:
10        print(math.factorial(i))
11
12    print(math.pi)
13    print(math.pi * 2)
14
15 do_some_math()
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

PORTS

SQL CONSOLE

TERMINAL

```
~/Desktop/Using Libraries
> python3 math_demo.py
5
1
2
6
24
120
720
5040
40320
362880
3.141592653589793
6.283185307179586

~/Desktop/Using Libraries
>
```

OUTLINE

TIMELINE

EXPLORER

USING LIBRARIES

dates_demo.py

math_demo.py

matplotlib_demo.py

random_demo.py

math_demo.py

dates_demo.py

dates_demo.py

do_datetime_stuff

```
1 import datetime
2
3 def do_datetime_stuff():
4     # get the current time
5     now = datetime.datetime.now()
6     print(now)
7
8     # get the current time (formatted)
9     now_formatted = now.strftime("%Y-%m-%d %H:%M:%S")
10    print(now_formatted)
11
12    # get the current date
13    today = datetime.date.today()
14    print(today)
15
16    yesterday = today - datetime.timedelta(days=1)
17    print(yesterday)
18
19 do_datetime_stuff()
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

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SQL CONSOLE

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~~~~~

NameError: name 'math' is not defined. Did you forget to import 'math'?

~/Desktop/Using Libraries

python3 math\_demo.py

5

1

2

6

24

120

720

5040

40320

362880

3.141592653589793

6.283185307179586

~/Desktop/Using Libraries

>

OUTLINE

TIMELINE

Launchpad 0 0 0 0 0 Connect

Ln 6, Col 15 Spaces: 4 UTF-8 LF Python 3.12.2 64-bit

EXPLORER

USING LIBRARIES

dates\_demo.py

math\_demo.py

matplotlib\_demo.py

random\_demo.py

math\_demo.py

dates\_demo.py

random\_demo.py

random\_demo.py

do\_random\_stuff

a\_random\_string

```
1 import random
2
3 def do_random_stuff():
4     a_random_number = random.random()
5     print(a_random_number)
6
7     a_random_integer = random.randint(1, 100)
8     print(a_random_integer)
9
10    options = ["Dune", "Foundation", "The Three-Body Problem", "Hyperion"]
11    a_random_string = random.choice(options)
12    print(a_random_string)
13
14 do_random_stuff()
```

PROBLEMS

OUTPUT

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TERMINAL

~/Desktop/Using Libraries

python3 random\_demo.py

0.07605978904766082

5

Dune

~/Desktop/Using Libraries

python3 random\_demo.py

0.02548552810271376

64

Foundation

~/Desktop/Using Libraries

python3 random\_demo.py

0.9742189195649643

41

Hyperion

~/Desktop/Using Libraries

python3 random\_demo.py

Ln 11, Col 20 · Spaces: 4 · UTF-8 · LF · Python · 3.12.2 64-bit

EXPLORER

USING LIBRARIES

dates\_demo.py

math\_demo.py

matplotlib\_demo.py

my\_first\_graph.png

random\_demo.py

math\_demo.py

dates\_demo.py

random\_demo.py

matplotlib\_demo.py

my\_first\_graph.png

matplotlib\_demo.py > make\_a\_graph

```
1 import matplotlib.pyplot as plt
2
3 def make_a_graph():
4     nums = []
5     squares = []
6     doubles = []
7
8     for i in range(-15, 15):
9         nums.append(i)
10        squares.append(i * i)
11        doubles.append(i * 2)
12
13    plt.plot(nums, squares)
14    plt.plot(nums, doubles)
15    plt.legend(['y = x^2', 'y = 2x'])
16    plt.savefig('my_first_graph.png')
17
18 make_a_graph()
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

PORTS

SQL CONSOLE

TERMINAL

Requirement already satisfied: fonttools>=4.22.0 in /Users/brahm/.pyenv/versions/3.12.2/lib/python3.12/site-packages (from matplotlib) (4.53.0)

Requirement already satisfied: kiwisolver>=1.3.1 in /Users/brahm/.pyenv/versions/3.12.2/lib/python3.12/site-packages (from matplotlib) (1.4.5)

Requirement already satisfied: numpy>=1.23 in /Users/brahm/.pyenv/versions/3.12.2/lib/python3.12/site-packages (from matplotlib) (1.26.4)

Requirement already satisfied: packaging>=20.0 in /Users/brahm/.pyenv/versions/3.12.2/lib/python3.12/site-packages (from matplotlib) (24.0)

Requirement already satisfied: pillow>=8 in /Users/brahm/.pyenv/versions/3.12.2/lib/python3.12/site-packages (from matplotlib) (10.3.0)

Requirement already satisfied: pyparsing>=2.3.1 in /Users/brahm/.pyenv/versions/3.12.2/lib/python3.12/site-packages (from matplotlib) (3.1.2)

Requirement already satisfied: python-dateutil>=2.7 in /Users/brahm/.pyenv/versions/3.12.2/lib/python3.12/site-packages (from matplotlib) (2.9.0.post0)

Requirement already satisfied: six>=1.5 in /Users/brahm/.pyenv/versions/3.12.2/lib/python3.12/site-packages (from python-dateutil>=2.7->matplotlib) (1.16.0)

Using cached matplotlib-3.9.0-cp312-cp312-macosx\_11\_0\_arm64.whl (7.8 MB)

Installing collected packages: matplotlib

Successfully installed matplotlib-3.9.0

~/Desktop/Using Libraries

python3 matplotlib\_demo.py

~/Desktop/Using Libraries 1m 18s

python3 matplotlib\_demo.py

~/Desktop/Using Libraries

OUTLINE

TIMELINE

Launchpad 0 0 0 Connect

Ln 16, Col 4 (34 selected) Spaces: 4 UTF-8 LF Python 3.12.2 64-bit ("3

EXPLORER

USING LIBRARIES

dates\_demo.py

math\_demo.py

matplotlib\_demo.py

my\_first\_graph.png

random\_demo.py

math\_demo.py

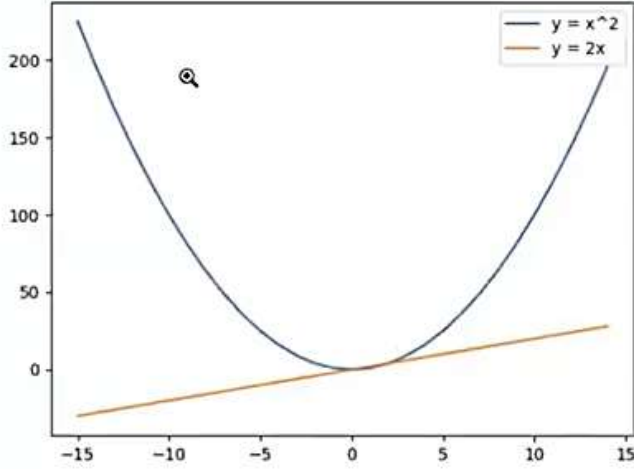
dates\_demo.py

random\_demo.py

matplotlib\_demo.py

my\_first\_graph.png X

my\_first\_graph.png



PROBLEMS

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TERMINAL

```
Requirement already satisfied: fonttools>=4.22.0 in /Users/brahm/.pyenv/versions/3.12.2/lib/python3.12/site-packages (from matplotlib) (4.53.0)
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Using cached matplotlib-3.9.0-cp312-macosx_11_0_arm64.whl (7.8 MB)
Installing collected packages: matplotlib
Successfully installed matplotlib-3.9.0

~/Desktop/Using Libraries
• › python3 matplotlib_demo.py

~/Desktop/Using Libraries 1m 18s
• › python3 matplotlib_demo.py

~/Desktop/Using Libraries
○ ›
```

OUTLINE

TIMELINE